C-MIL-T-21671(NOrd)

18 November 1958
SUPERSEDING
NAVORD OS 5869B
9 March 1955

MILITARY SPECIFICATION

TEST SET MARK 126 MOD 0

1. SCOPE

1.1 This specification establishes the requirements for the procurement of the Test Set Mark 126 Mod 0. The test set is used to check resistor and capacitor tolerances, continuity, and insulation resistance of Sensitivity Switch Mark 5 Mods 0 and 1.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Boxes, Fiber.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-131	Barrier Material; Water Vapor-proof, Flexible.
MIL-V-173	Varnish, Moisture-and Fungus Resistant for the Treatment of Communications, Electronic and Associated Electrical Equipment.

FSC 6625

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MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification.
MIL-S-6872	Soldering Processes, General Specifications for.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

BUREAU OF ORDNANCE

LD 274820	Test Set Mark 126 Mod 0.
LD 284866	Preparation for Delivery for Test Set Cable Assemblies.
1476194	Marking Instructions for Storage and Shipping Containers.

And all documents listed in the above LD's and drawing.

PUBLICATIONS

BUREAU OF ORDNANCE

OP 400	General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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3. REQUIREMENTS

3.1 General requirements.

3.1.1 The Test Set Mark 126 Mod 0 shall be manufactured in accordance with LD 274820 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be complied with as applicable.

3.1.3 Preproduction sample. - Unless otherwise specified in the contract or order, a preproduction sample shall be manufactured by the same procedures and processes and at the same location as proposed for full production. This sample will be tested for the purpose of determining complicance with the requirements of this specification, the drawings, and the contract. See 4.1.2.

3.1.4 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior approval of the Bureau of Ordnance.

3.2 Moisture-and fungus-resistant treatment. - Materials and parts of this test set which are not moisture and fungus resistant shall be treated, before assembly, with a varnish conforming to Military Specification MIL-V-173.

3.3 Requests for deviation. - Requests for deviation from this specification, applicable drawings, specifications, publications, materials and processes specified herein, shall be as set forth in OP 400.

3.4 Conflicting requirements. - Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

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3.5 Soldering. - Unless otherwise specified, soldering processes shall conform to the requirements specified in Military Specification MIL-S-6872. Only non-corrosive and non-volatile fluxes shall be used. Rosin shall be used when soldering wires to contacts and terminals. The use of rosin core solder is permitted. Electrical connections shall be mechanically secure and electrically continuous before soldering. Sufficient heat shall be applied during the soldering process to avoid "cold" solder joints.

3.6 Wiring. - All internal wiring shall be cabled wherever possible.

3.7 Performance requirements and product characteristics. - The test set covered by this specification shall meet the following performance requirements and product characteristics.

3.7.1 Insulation resistance. - The insulation resistance between electrical circuits and between each circuit and the chassis shall be not less than 50 megohms when tested as specified in 4.5.3.

3.7.2 Calibration and percentage control. - The calibration and percentage controls shall adjust the "magic eye" tube to within the specified limits, when tested as specified in 4.5.4.

3.7.3 ADAPTER CABLE connector. - With the components that are supplied with the test set connected between ADAPTER CABLE pins and binding posts indicated, the percentage control shall indicate 0 ±2 percent when tested as specified in 4.5.5.

3.7.4 SELECTOR switch. - An external ohmmeter shall read as specified when the SELECTOR switch circuitry is tested as specified in 4.5.6.

3.7.5 SENS. SW. CAP. connector. - An external ohmmeter shall read as specified when the SENS. SW. CAP. circuitry is tested as specified in 4.5.7.

3.8 Workmanship and cleanliness. - The test set shall be free from dust, dirt, metal filings, moisture, solder splashes, chips, and other foreign matter. Necessary precautions shall be taken during manufacture and assembly to insure these conditions.

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4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS.

4.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of MIL-STD-105.

4.1.1 Lot. - As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment or other purpose.

4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of one test set shall be manufactured using the methods and procedures proposed for full production. The Government inspector shall perform, witness, or participate in such inspections and tests on the sample as are required for full production. Upon conditional acceptance by the inspector, the preproduction sample shall be forwarded, together with a report of inspection, to an activity designated in the contract or by the procuring agency, for the purpose of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of the tests on preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.2.1 Preproduction sample for a subsequent contract. - The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this document.

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4.2 Inspection.

4.2.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that all components and assemblies are in accordance with the requirements of the contract, pertinent specifications, and drawings. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.2.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspection and testing of test sets covered by this document shall be classified as follows:

(a) preproduction tests

(b) acceptance tests

4.3.1 Preproduction tests. - Preproduction tests are those performed on samples submitted in accordance with 4.1.2 to determine that the lot and production meet all the requirements of this specification.

4.3.2 Acceptance tests. - Acceptance tests for acceptance of the test set covered by this specification shall consist of tests to determine compliance with all requirements specified in this document.

4.4 Test equipment.

4.4.1 Test and inspection equipment and facilities. - Unless otherwise specified, the manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests under the supervision of the Government inspector. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

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4.4.1.1 Equipment required for performing these tests is as follows:

(a) 115-volt, 60 cycle, a-c supply.

(b) Ohmmeter, ranges 0 to 5, 1000, 5000, 1000,000, and 500,000 ohms, accuracy within ±2 percent.

(c) Megohmmeter, 500-volts d-c, range 10.0 to 100 megohms, accuracy within ±10 percent.

4.5 Tests.

4.5.1 Test conditions. - Unless otherwise specified, tests shall be performed at 80° ±10°F and ambient relative humidity.

4.5.2 Visual inspection. - The test set and its component parts shall be inspected to verify that the materials, design, construction, physical dimensions, soldering, and workmanship are in accordance with this document and the drawings.

4.5.3 Insulation resistance. - The test set shall be subjected to a conditioning period of 24 hours at a temperature of 77° ±2°F, and a relative humidity or 75 ±5 percent. A test potential of 500 ±20 volts d-c is then applied as specified in 3.7.1, using megohmmeter, test item 4.4.1.1 (c). The insulation resistance shall be not less than 50 megohms.

4.5.4 Calibration and percentage control test.

(a) Connect test set AC plug to 115-volt, 60-cycle power supply, test item 4.4.1.1 (a).

(b) Set PWR switch at ON and allow 5 minutes for the set to warm up.

(c) Connect one 20,000-ohm resistor (supplied with set) to STANDARD binding posts.

(d) Connect other 20,000-ohm resistor (supplied with set) to UNKNOWN binding posts.

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(e) Set SELECTOR switch at RES.-COND.

(f) Turn percentage control until minimum opening is observed on "magic eye" tube. Opening width shall be between 1/16-inch and 1/32-inch (obtainable with EYE ADJ. control). Loosen setscrew on percentage control knob and set knob at zero. Tighten setscrew.

(g) Remove resistors from binding posts.

(h) Connect 50-megohm resistor (supplied with set) to J-N binding posts.

(i) Set SELECTOR switch at LKG. J-N.

(j) Turn LKG. BAL control until minimum opening is observed in "magic eye" tube. Opening width shall be between 1/16-inch and 1/8-inch, and shall be obtainable with BRG.SENS. control.

(k) Remove resistor from binding post.

(l) Set PWR switch at OFF and disconnect test set from AC power supply.

4.5.5 ADAPTER CABLE connector test.

(a) Connect test set to 115-volt, 60-cycle power supply, test item 4.4.1.1 (a), with CA-731.

(b) Set PWR switch at ON and allow 5 minutes for set to warm up.

(c) Connect components (supplied with test set) between ADAPTER CABLE pins and binding post as shown below.

(d) With SELECTOR switch at positions listed, percentage control shall indicate 0 ±2% when "magic eye" tube indicates minimum opening.

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SELECTOR Switch Setting	Component	ADAPTER CABLE	BINDING POST
A-N	20,000 ohms	A	N
B-N	10,000 ohms	B	N
C-N	6,000 ohms	C	N
F-N	51,000 ohms	F	N
G-N	1.0 mfd.	G	J

(e) Disconnet test set from 115-volt supply. Ohmmeter shall indicate direct connection between pins J, E, and D of ADAPTER CABLE.

4.5.6 SELECTOR switch test.

(a) Connect ohmmeter, test item 4.4.1.1 (b), to pin G of SENS. SW. CAP. connector and pin L of ADAPTER CABLE connector.

(b) With SELECTOR switch set at position listed, ohmmeter shall indicate values shown in table, ±10 percent.

SELECTOR Switch Setting	Ohmmeter indication (ohms)
A-N	1935
B-N	1460
C-N	1460
F-N	3150
G-N	1475
G-A	42,000-52,000 (adjustable with percent control)
RES. COND. INS. RES.) LKG. J-N) LKG. DET.)	1475 Adjustable from 0 to 500,000 ohms with BRG. SENS. control.

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(c) Set SELECTOR switch at last (not engraved) position. Ohmmeter shall indicate open circuit.

(d) Remove ohmmeter.

4.5.7 SENS. SW. CAP. connector test.

(a) Connect ohmmeter, test item 4.4.1.1 (b), to pins listed and set SELECTOR switch as indicated. Ohmmeter shall indicate as shown in following table:

Ohmmeter Pin Connections

SELECTOR Switch Setting	ADAPTER CABLE	SENS. SW. CAP.	Ohmmeter Indication
G-A	L	A	540 ohms ±10%
G-A		B-F	direct connection
G-A		C-F	direct connection
G-A		D-F	direct connection
G-A		E-F	direct connection
INS. RES.	J	F	direct connection

4.6 Criteria for acceptance.

4.6.1 Preproduction tests. - Acceptance of preproduction sample shall be based on no defects in the sample.

4.6.2 Acceptance tests. - Failure of any test set to pass the acceptance tests shall result in rejection of the test set. If the number of failing test sets exceeds the number permitted by the associated criteria, the lot represented will be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning and drying. - (Not applicable).

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5.1.1.2 Preservative application. - (Not applicable).

5.1.1.3 Unit packaging. - Prior to unit packaging, test set cables shall be packaged in accordance with Bureau of Ordnance List of Drawings LD 284866 and positioned in cable well of test set. Each test set shall be initially packaged in a Style RSC fiberboard box fabricated from Class 1, Type I or Type II board, having 275 lb. minimum bursting strength, conforming to the requirements of Federal Specification PPP-B-636. Sufficient fillers and folded corner pads, of the same box material, shall be utilized within the carton to provide a tight pack and to prevent direct contact between protrusions of the test set case and the walls of the fiberboard box.

5.1.1.3.1 Each test set, prepared as specified above, shall be further packaged Method II-b conforming to the requirements of Military Specification MIL-P-116. The inner carton, containing the packaged test set, shall be inserted into a water-vaporproof bag, fabricated from Class 1 barrier-material conforming to requirements of Military Specification MIL-B-131, and closure heat-sealed.

5.1.1.3.2 A style RSC fiberboard box, Class 2, W5c or W5s compliance symbol, conforming to the requirements of Federal Specification PPP-B-636, shall be employed for the outer carton and designed to snugly enclose the barrier-sealed inner container. The required desiccant shall be suitably positioned within the test set case. For conformance and quantity required see Military Specification MIL-D-3464.

5.1.2 Level B.

5.1.2.1 Cleaning and drying. - (Not applicable).

5.1.2.2 Preservative application. - (Not applicable).

5.1.2.3 Unit packaging. - Test set shall be prepared and packaged as specified under 5.1.1.3.

5.1.2.3.1 Each test set, prepared as specified above, shall be further packaged Method IA-14, conforming to the requirements of Military Specification MIL-P-116. The inner carton, containing the packaged test set, shall be inserted into a water-vaporproof bag, fabricated from Class I barrier-material conforming to requirements of Military Specification MIL-B-131, and closure heat-sealed.

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5.1.2.3.2 A Style RSC fiberboard box, Class 2, W5c or W5s compliance symbol, conforming to the requirements of Federal Specification PPP-B-636, shall be employed for the outer carton and designed to snugly enclose the barrier-sealed inner container. The required desiccant shall be suitably positioned within the test set case.

5.1.3 Level C.

5.1.3.1 Test set units shall be packaged in a manner which will afford adequate protection from deterioration and physical damage during shipment from the contractor or supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior containers. - Each test set, individually packaged as specified under 5.1.1, shall be packed in an overseas type wood-cleated plywood box or a Class 2, Style 2, 2-1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specification PPP-B-601 or PPP-B-621, respectively.

5.2.2 Level B.

5.2.2.1 Exterior containers. - Each test set, individually packaged as specified under 5.1.2, shall be packed in a domestic type wood-cleated plywood box or a Class 1, Style 2, 2-1/2, 3 or 4 nailed wood box conforming to the requirements of Federal Specification PPP-B-601 or PPP-B-621, respectively.

5.2.3 Level C.

5.2.3.1 Exterior containers. - Each test set shall be packed in a manner which will provide protection from damage during direct shipment from the contractor or supply source to the first receiving activity for immediate use. The type of containers used shall conform to applicable carrier rules and regulations.

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5.3 Marking.

5.3.1 Normal markings. - In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129.

5.3.1.1 For details on marking requirements, see Bureau of Ordnance Drawing Number 1476194.

5.3.1.2 Special markings. - The following special marking shall appear on the top, one (1) side and one (1) end of each shipping container:

DELICATE INSTRUMENT
HANDLE WITH CARE

6. NOTES.

6.1 Ordering data. - Procurement documents should specify the title, number and date of this specification, and exceptions to this specification, applicable drawings and other documents. Procurement documents should also specify the desired level of packaging and packing. Level C should not be selected unless it is definitely known that the equipment will be used immediately upon receipt.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever: and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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